
Structure of a Recursive Function:

A recursive function typically has two main components:

1. **Base Case:** The condition under which the function stops calling itself.
2. **Recursive Case:** The function calls itself with a modified argument.

Example: Factorial Calculation

```
def factorial(n):
    if n == 1:  # Base case
        return 1
    else:  # Recursive case
        return n * factorial(n - 1)

print(factorial(5))  # Outputs: 120
```

Explanation:

- **Base Case:** `if n == 1: return 1` stops the recursion.
- **Recursive Case:** `return n * factorial(n - 1)` multiplies `n` by the factorial of `n-1`.

Applications of Recursion: